Better Movement Better Life

MEDICOS MINDS

Study Smart
Move Better
Live Healthy

THE SCIENCE OF DRUGS

PHARMACOLOGY

ESSENTIALS FOR ALL MEDICAL STUDENTS

Understand Drugs. Improve Care. Save Lives.





Right Drug
Right Dose
Right Patient
Right Time

KEY TOPICS

- Drug Classification & Pharmacokinetics
- Pharmacodynamics
- Therapeutic Uses
- Adverse Drug Reactions
- Drug Interactions
- Contraindications
- Dose, Routes & Formulations
- Toxicology & Management

WHY PHARMACOLOGY?

- Essential for rational drug use
- Helps in disease management
- Prevents drug interactions
- Improves patient safety
- Builds clinical decision making

From Molecule to Medicine



REMEMBER

- Small dose Big impact.
- Drugs don't work in isolation.

THIS NOTE IS VALUABLE FOR

MBBS | BHMS | BDS | BSc.NURSING | BPT

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Pharmacology Builds Better Therapies

SAME BOOK | ALL MEDICAL STUDENTS | ONE GOAL | BETTER HEALTHCARE

PHARMACOLOGY | MEDICOS MINDS

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HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Pharmacology

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OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

Format: High-Yield Digital PDF | Standard Price: Rs. 99

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

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PHARMACOLOGY

2. ROUTES OF DRUG ADMINISTRATION, DISTRIBUTION, METABOLISM & EXCRETION

1) ROUTES OF DRUG ADMINISTRATION

A. ENTERAL ROUTES

1. Oral (Per oral)
   - Most common & convenient.
   - Slow onset; affected by food, pH & first-pass metabolism. Ex: Paracetamol
2. Sublingual / Buccal
   - Drug absorbed through oral mucosa.
   - Bypasses first-pass metabolism; rapid onset. Ex: Nitroglycerin
3. Rectal
   - Useful in vomiting, unconscious or pediatrics.
   - Partial first-pass avoidance. Ex: Diclofenac suppository

B. PARENTERAL ROUTES

1. Intravenous (IV)
   - 100% bioavailability; immediate effect.
   - Used in emergencies, fluids, irritant drugs.
2. Intramuscular (IM)
   - Absorbed by muscle blood flow.
   - Moderate onset; suitable for oily solutions. Ex: Diclofenac inj.
3. Subcutaneous (SC)
   - Absorbed slowly; for small volumes.
   - Ex: Insulin, Heparin
4. Intradermal (ID)
   - Into dermis layer; very small volume.
   - Used for allergy tests, TB test (PPD).

C. TOPICAL / TRANSDERMAL

- Applied on skin or patches.
- Local effect with minimal systemic effect.
- Ex: Diclofenac gel, Fentanyl patch, Nicotine patch

D. INHALATIONAL

- Drug delivered via lungs.
- Rapid absorption, large surface area.
- Ex: Salbutamol MDI, Anesthetics

2) DRUG DISTRIBUTION

Definition: Reversible process by which a drug moves from bloodstream to tissues & body fluids.

★ Plasma Protein Binding
- Drugs bind to albumin, α_1-acid glycoprotein.
- Only free (unbound) drug is pharmacologically active & able to cross membranes.

★ Factors Affecting Distribution
- Lipid solubility (↑ lipid solubility → ↑ distribution)
- Plasma protein binding
- Tissue perfusion
- Membrane permeability
- Molecular size & ionization (pH dependent)
- Body composition (↑ fat → ↑ volume of distribution)

★ Blood-Brain Barrier (BBB)
- Selectively permeable; protects CNS.
- Lipid soluble, unionized drugs cross easily.
- Ex: Thiopentone crosses; Heparin does not.

★ Placental Barrier
- Many drugs cross placenta to fetus.
- Lipid soluble, small, non-ionized drugs cross easily.
- Ex: Thalidomide (teratogenic), Heparin (does not).

3) METABOLISM / BIOTRANSFORMATION

- Mainly occurs in LIVER (hepatocytes).
- Converts lipophilic drugs to more water soluble metabolites for easy excretion.
- Two Phases:

Purpose:
- Detoxification
- Inactivation
- Facilitate excretion

CYP450 (Cytochrome P450) → Important enzyme system in Phase I.

A. Phase I (Functionalization)
- Oxidation, Reduction, Hydrolysis.
- Introduces/uncovers functional groups (-OH, -NH_2, -SH).
- Mostly via CYP450 enzyme system.
- Ex: Diazepam → Desmethyldiazepam (oxidation).

B. Phase II (Conjugation)
- Conjugation with endogenous substances.
- e.g., Glucuronic acid, Sulfate, Glycine, Acetate, Glutathione.
- Produces highly water soluble, inactive metabolites.
- Ex: Paracetamol + Glucuronic acid → Conjugate

4) EXCRETION

1. Kidneys (Urine) – Major route
   - Glomerular Filtration: Free drug filtered.
   - Tubular Secretion: Active secretion into tubule.
   - Tubular Reabsorption: Lipid soluble, non-ionized drugs reabsorbed (Depends on urine pH).
   - pH trapping: Alkaline urine ↑ excretion of weak acids; Acidic urine ↑ excretion of weak bases.
2. Bile (Feces)
   - Drugs excreted in bile → intestine → feces. Ex: Rifampicin, Erythromycin
3. Lungs
   - Volatile substances exhaled in expired air.
   - Ex: Anesthetic gases, Alcohol
4. Sweat
   - Small amounts excreted in sweat.
   - Ex: Salicylates
5. Saliva & Milk
   - Small amounts appear in saliva & breast milk.
   - Important in nursing mothers.

EXAM PEARLS

★ First-Pass Effect: Metabolism of drug in liver after oral absorption → ↓ bioavailability.
★ Bioavailability (F): Fraction of unchanged drug reaching systemic circulation.
★ IV Route has 100% Bioavailability (F = 1).
★ Lipid soluble, non-ionized drugs → cross BBB & Placenta easily.
★ Phase I = Functionalization (CYP450), Phase II = Conjugation.

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THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

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